STRATEGIC CAPITAL PORTAL

FS-1 DETERMINISTIC AGRICULTURE OS

Technical Architecture

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Investor Material	CONFIDENTIAL	March 9, 2026

MASTER SOFTWARE ARCHITECTURE: FarmSense OS & Logic (V2.1)
The Single Source of Truth for Software Engines, APIs, and Data Flows Consolidates: ARCHITECTURE.md, BLUEPRINT.md, FEATURESET.md, BACKEND_SERVICE_MAP.md, IMPLEMENTATION_GUIDE.md, and 2 Software KIs.

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1. System Philosophy & Logic Mandate

FarmSense is a **Deterministic Farming Operating System**. It is engineered to replace stochastic "intuition" with a high-fidelity computational engine.

1.1 The Deterministic Mandate

No Black-Box AI: All irrigation and trading decisions are deterministic and judgment-based. This is a non-negotiable requirement for **Water Court Admissibility**.

Evidence-Grade Tech: Every packet is signed at the edge (**AES-256**), and every decision is logged in a SHA-256 hash-chained compliance ledger.

Explainable Logic: Operators must be able to audit why a "Soft-Stop" or "VRI-Update" was issued back to raw sensor telemetry.

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2. Tiered Lambda Architecture

FarmSense functions as a decentralized monolithic grid, balancing low-latency edge reflex with high-capacity cloud geostatistics.

2.1 Hierarchical Processing Stack

Level 1 (Field): LRZ1/LRZ2/VFA/PFA (LoRa bursts) → **PMT Hub** (50m Grid, Edge-EBK baseline). 900MHz **Chirp Spread Spectrum (CSS)** security.

Level 2 (District): DHUs (NVIDIA Jetson Orin Nano) → 1m/10m/20m Grid (Go-based Bayesian Kriging). Localized reflex autonomy.

Level 3 (Regional): RSS (64-Core Threadripper) → 1m Grid (Python-based Regression Kriging + FHE). Regional DIL vaulting.

Level 4 (Global): Core Compute Cloud → Multi-field analytics, Federated Learning, global hydro-economics.

```
graph TD
    subgraph "Level 4: Global Cloud"
        Cloud[Core Compute Cloud]
    end

    subgraph "Level 3: Regional"
        RSS[Regional Superstation - Threadripper]
    end

    subgraph "Level 2: District"
        DHU[District Hub - Jetson Orin]
    end

    subgraph "Level 1.5: Field Hub"
        PMT[Pivot Motion Tracker - ESP32]
    end

    subgraph "Level 1: Source Nodes"
        VFA[Vertical Field Anchor]
        PFA[Pressure & Flow Anchor]
        LRZ[Lateral Root Zone]
```

```
    end

    VFA -- "LoRa (900MHz)" ⟶ PMT
    PFA -- "LoRa (900MHz)" ⟶ PMT
    LRZ -- "LoRa (900MHz)" ⟶ PMT

    PMT -- "2.4GHz / LTE-M" ⟶ DHU
    DHU -- "Fiber / LTE-M" ⟶ RSS
    RSS -- "Fiber" ⟶ Cloud

    Cloud -. "Federated Learning" .→ RSS
    RSS -. "Kriging Matrix" .→ DHU
    DHU -. "VRI Worksheet" .→ PMT
```

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2.2 Core Software Tier Stack

Layer	Node	Engine Grade	Role
L0 (Field)	Sensors	Bare-Metal C	Raw sensing & local deep-sleep management.
L1 (Hub)	PMT	ESP32-S3	Reflex Logic (IMU Stall → Stop), 50m Grid Compute.
L2 (District)	DHU	Orin Nano / Go	20m/10m Grid Kriging, AllianceChain PBFT Consensus.
L3 (Core Compute Engine)	RSS / Cloud	Python / FastAPI	1m Master Grid, Long-term DIL, Legal Vault.

3. Data Infrastructure & Schemas

3.1 TimescaleDB (Time-Series Telemetry)

Standardized monthly chunking for high-velocity sensor data.

```
CREATE TABLE sensor_readings (
    time          TIMESTAMPTZ NOT NULL,
    device_id     UUID NOT NULL,
    field_id      UUID NOT NULL,
    sensor_type   VARCHAR(50), -- 'moisture_10cm', 'moisture_30cm', 'flow_rate'
    value         DOUBLE PRECISION,
    quality_score FLOAT,       -- 0.0 to 1.0 confidence
    metadata      JSONB,       -- RSSI, BatVoltage, ADC_Raw
    PRIMARY KEY (time, device_id, sensor_type)
);
SELECT create_hypertable('sensor_readings', 'time');
```

3.2 PostgreSQL (Compliance & Ledger)

Implements blockchain-style hash chaining for tamper-proof regulatory reporting.

```
CREATE TABLE compliance_logs (
    id            UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    field_id      UUID REFERENCES fields(id),
    log_time      TIMESTAMPTZ NOT NULL DEFAULT NOW(),
    event_type    VARCHAR(50), -- 'IRRIGATION_EVENT', 'VIOLATION', 'PBFT_COMMIT'
    details       JSONB NOT NULL,
    hash          VARCHAR(64), -- SHA-256 of (details + previous_hash)
    previous_hash VARCHAR(64)
);
```

3.3 TimescaleDB Table: `soil_sensor_readings`

(Hypertable chunked by 7 days)

Column	Type	Index
`sensor_id`	`UUID`	`BTREE`
`timestamp`	`TIMESTAMPTZ`	**Clustered**

Column	Type	Index
`field_id`	`VARCHAR(32)`	`HASH`
`moisture_surface`	`FLOAT4`	—
`moisture_root`	`FLOAT4`	—
`location`	`GEOGRAPHY(POINT)`	`GIST`

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4. Core Software Engines

4.1 The Core Compute Engine (Soil-Plant-Atmosphere Continuum Synthesis)

Hosted at `brodiblanco.zo.computer`. Responsible for the **Soil-Plant-Atmosphere Continuum (SPAC)** synthesis and high-fidelity moisture mastering.

Bayesian Priors: Establishes moisture probability using historical **Soil Functional Domain (SFD)** profiles and multispectral NDVI baselines.

Regression Kriging: Ordinary Kriging corrects Sentinel-2 trend bias using ground-truth LRZ1/LRZ2/VFA **Dielectric Telemetry**.

MAD Framework: **Management Allowable Depletion (MAD)** — utilizes the soil profile as a "Water Battery" to optimize pumping schedules and prevent permanent wilting points.

4.2 Adaptive Recalculation Engine ("Fisherman's Attention")

The PMT continuously executes **Edge-EBK** to generate a 50m-resolution spatial probability grid (16×16 matrix). The execution frequency is dynamically governed by environmental volatility:

Mode	Trigger Threshold	Logic Execution Frequency
Dormant	Stable soil moisture + Pivot Parked	Every 4 Hours
Anticipatory	Sunrise / $T > 5^\circ$C rise/hr	Every 60 Minutes
Focus Ripple	Anomaly detected (>5% deviation)	Every 15 Minutes
Focus Collapse	Mainline Pressure > 1 PSI / Tower Motion	Every 5 Seconds

Autonomous Edge-EBK Logic

FPU Calculation: The hardware FPU processes **AES-256** chirps from LRZ1/LRZ2/VFA nodes into a localized 16×16 probability matrix.

Trajectory Collapse: In "Collapse" mode, the FPU zeroes calculations on dormant sections, focusing 100% of compute on the active pivot span trajectory.

Payload Bundling: The PMT bundles its own High-Fidelity kinematic data, the processed 50m Edge-EBK arrays, and the intercepted VFA/LRZ intelligence into a unified, encrypted **~187-byte Field State Payload**.

LoRa Mesh Backhaul: Blasts the unified payload to the District Hub (DHU) via 900MHz LoRa Mesh.

Zero-Downtime VRI Failover: Upon loss of DHU mesh-ping, the PMT instantly executes autonomous Variable Rate Irrigation based onlocalized intelligence.

Aggregate Harmonic Analysis: Ingests raw current wave descriptors from the **PFA (Level 1)** and executes **Fast Fourier Transform (FFT)** via vector instructions to detect motor cavitation or bearing failure.

Audit Buffering: Stores all payload state changes to onboard SPI Flash, burst-transmitting the backlog upon reconnection to preserve the State Engineer audit ledger.

4.3 Level 1 "Dumb Chirp" Transformation (VFA & PFA)

The Level 1 "Source" nodes (VFA/PFA/LRZ) are optimized for a 10-year battery life and unified silicon (nRF52840):

AES-256 Encryption at the Edge: Independently encrypts the localized payload (Soil Moisture or Wellhead Waveform) before transmission.

Dumb Chirp Mode: Minimizes CPU cycles by transmitting a fixed-length encrypted burst to the overhead PMT Field Hub.

Dynamic Ripple Scaling: Responds to PMT "Ripple" pings, scaling chirp frequency during detected anomalies.

LPI/LPD Constraints: Firmware ensures **Chirp Spread Spectrum (CSS)** conforms to Federal Low Probability of Intercept/Detection standards.

Volatility Score Logic (Decision Engine)

`Volatility = (Moisture_Δ_1h * 0.4) + (Irrigation_Active * 0.3) + (VPD_Stress * 0.2) + (Wind_Stress * 0.1)`

Score > 0.7 → COLLAPSE

Score > 0.3 → ANTICIPATORY

Default → DORMANT

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4.3 Decision Engine (Reflex Logic)

Sub-cloud orchestrator evaluation field-state conditions.

`PMT_STALL` → Issue `ACTUATE_STOP` to PFA Relay.

`LINE_PRESSURE_MAX` → Issue `REFLEX_STOP`.

`NOZZLE_DROP` → Trigger VRI recalculation at 1m.

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5. Backend Service Map (`app/services/`)

Service Module	Function	Algorithm/Tech
`rss_kriging.py`	1m Mastering	Scikit-learn GPR (RBF + WhiteKernel) + Spatial Prior Fusion.
`csa_alignment.py`	Corner Resolver	Law of Cosines Kinematic Handshake resolving joint elbow to ±0.1°.
`trading_service.py`	Water Market	PBFT AllianceChain Consensus on DHU Industrial SSDs.
`predictive_maint.py`	Pump Health	Current Harmonic Analysis (NXP M7) detecting torque ripple & cavitation.
`spatial_privacy.py`	Data Obfuscation	Contextual Anonymization segregating Legal Ledger from Global Analytics.
`globalGAP_compliance`	Certification	Automated 1-click sustainability audit trail generation.
`jadc2_adapter.py`	Inter-agency	JADC2 / LPI / LPD tactical environmental tactical translation.
`satellite_service`	NDVI/NDWI	Sentinel-2 / Landsat-9 Multi-temporal stack with atmospheric correction.
`decision_engine.py`	Reflex Logic	Deterministic rule-based boolean actuation (MAD Framework).

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6. Security, Authentication & Roles

6.1 Authentication (JWT Flow)

Backend: FastAPI + Bcrypt (Cost Rounds = 12).

Storage: `sessionStorage` (Stateless).

Secret: Managed via `JWT_SECRET` environment variable.

6.2 Role-Based Access Control (RBAC)

INTERNAL: Full system admin & OTA management.

FARMER: Own field telemetry, VRI command center, trading dashboard.

REGULATOR: Compliance audit viewer, non-repudiable ledger access.

INVESTOR: Regional equity scores, depletion risk, Water-ROI analytics.

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7. Strategic Feature Set

Resolution Pop: Dynamic UI blur/unlock between 50m (Free) and 1m (Enterprise) spatial tiers.

AllianceChain: Peer-to-peer water credit trading rights executed via DHU PBFT consensus.

Dual-Layer Contextual Anonymization: Segregates Hyper-accurate Legal Ledger (Private) from Federated Learning Models (Public/Aggregate).

Rapid Deployment/Dual-Use: JADC2 adapters translate soil trafficability into tactical terrain maps.

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8. 20-Week Implementation Roadmap (SLV Pilot)

Phase 1: Foundation (Weeks 1-4)

Week 1-2: Deploy unified CSE stack on `brodiblanco.zo.computer`. Setup Monitoring (Prometheus/Grafana).

Week 3-4: Implement JWT Auth & core SQLAlchemy models for sensor readings.

Phase 2: Ingestion & Kriging (Weeks 5-8)
Week 5: Implement high-velocity Sensor Ingestion API (Target: 10K pings/sec).
Week 6-8: Deploy 20m Edge Kriging on Jetson Hubs. Integrate Sentinel-2 cloud-filtering (30% threshold).
Phase 3: Analytics & Reflex (Weeks 9-12)
Week 9-10: Adaptive Recalculation Engine validation (Dormant → Collapse).
Week 11-12: Decision Engine "Soft-Stop" field trials.
Phase 4: Portals & Compliance (Weeks 13-16)
Week 13-14: Farmer Dashboard (React 18 + MapLibre GL JS).
Week 15: Regulatory Portal for SLV 2026 Water Court hearings.
Week 16: Security penetration testing & UAT with pilot farmers.
Phase 5: Optimization & Rollout (Weeks 17-20)
Week 17-18: Performance tuning (Kriging < 5 min/field).
Week 19: Trainer-the-trainer manual generation.
Week 20: **CSU SLV Research Center Pilot Launch** (2 Center Pivots, Full Stack).

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9. Performance & Health Targets

Metric	Target	Critical Threshold
API Response (p95)	< 200ms	500ms
20m Grid Calc	< 30 sec	60 sec
1m Grid Master	< 5 min	10 min
Sensor Ingestion	10,000 / sec	5,000 / sec
System Uptime	99.9%	99.5%

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10. CI/CD & Deployment Strategy
Automated deployment is managed via GitHub Actions to the CSE/Zo stack.
10.1 Environment Configuration
RDC (Oracle): Public IP for Map Database. Requires `POSTGRES_USER`, `POSTGRES_PASSWORD`.
Core Compute Engine: Core Application Server. Requires `MAP_DATABASE_URL` pointing to the RDC.
10.2 Deployment Workflow
Push to `main`: Triggers GitHub Action.
Step 1 (Validate): Lints and unit tests (Pytest/Jest).
Step 2 (Deploy-RDC): Deploys `docker-compose.oracle.yml` to the map stack.
Step 3 (Deploy-Zo): Deploys `docker-compose.zo.yml` to the core platform.

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11. AI Agent Coordination (SOP)
This section contains critical invariants for AI Assistants (antigravity/cortex) operating within the FarmSense codebase.
11.1 The "Before Creating" Rule
Before creating ANY new file, route, or service:
Audit Documentation: Start at `README.md` → 4 Masters.
Verify Routes: Check `app/api/main.py` for existing endpoint definitions.
Grep Search: Sweep the workspace for similar logic/classes. **Assumption is always that it already exists.**
11.2 Core Invariants
No Dependencies: Do not add external Python or JS libraries without explicit architectural review.
Deterministic Logic: Never propose or implement black-box ML for irrigation or ledger decision points.
Version Integrity: Maintain the Master Documents as the *only* authoritative technical specifications. Any engineering change must be reflected in the relevant Master first.
12. Hybrid Cloud Deployment (Zo + RDC)
FarmSense utilizes a split-deployment architecture to separate compute-heavy processing from geospatial data services.

12.1 RDC Cloud (Map Stack)

Hosts the PostGIS database and tile-service engine.

Port 5432: Database access.

Port 8001: Tile serving.

Path: `farmsense-code/deployment/docker/docker-compose.rdc.yml`

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12.2 Zo Server (Core Platform)

Hosts the FastAPI engine, React frontend, and Audit ledger.

Environment: `MAP_DATABASE_URL` must point to the RDC instance.

Path: `farmsense-code/deployment/docker/docker-compose.cse.yml`

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